

Sai Raman

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Profile

Experienced in building real-time analytics systems and scalable ML pipelines. Passionate about developing data-driven solutions, extracting actionable insights, and applying machine learning for optimization and intelligent decision-making in real-world systems.

Education

BTech Computer Science and Engineering with Artificial Intelligence

08/2023 – 05/2027

Amrita Vishwa Vidyapeetham, Chennai

Chennai, India

current cgpa: 8.47 / 10 (Till 5th semester)

Relevant Experience

Hack-A-Ruckus Event (An Amrita Vishwa Vidyapeetham event, Chennai) 

Led technical coordination for a college hackathon with 100+ participants and developed a responsive event website deployed on Vercel to manage registrations and information.

Skills

Programming Languages: Python, Java.

Data Science & AI: Data Analysis, Hypothesis Testing, Statistics, Regression Modelling, Machine Learning, Time-Series Analysis, Exploratory Data Analysis (EDA), Deep Learning, Generative AI, Natural Language Processing, Image Processing (OpenCV).

Software Development (SDE): Data Structures & Algorithms (Competitive Programming), Object-Oriented Programming (OOP), Backend Development (FastAPI), API Design & Integration, Database Management, Debugging, Version Control (Git-based workflows), Linux (Basics), AWS (Basics)

Tools & Frameworks: MySQL, MongoDB, Supabase, Firebase, Hadoop, Spark, Kafka, Tableau, Git, GitHub, Jenkins, Docker.

Soft Skills: Problem-Solving, Analytical Thinking, Team Collaboration, Communication, Adaptability.

Projects

Real-Time Traffic Optimization using OpenCV

Designed and implemented a smart traffic control system using OpenCV and lane-specific sensors to count vehicles in real time. The system detects congestion levels per lane and dynamically adjusts signal timings, prioritizing high-density lanes to improve traffic flow.

Datacure - Automated Dataset Curation & AutoML Platform

Built an end-to-end data analytics and AutoML platform using Streamlit for rapid prototyping of ML pipelines. Implemented automated data cleaning, feature engineering, and model selection workflows with real-time evaluation dashboards. Designed as a proof-of-concept system to accelerate experimentation and reduce manual effort in analytics pipelines.

Real-Time IoT Anomaly Detection using Apache Kafka and Hidden Markov Models

Developed a real-time industrial anomaly detection system using Apache Kafka and Hidden Markov Models (HMM) with Viterbi inference. Built a streaming pipeline to detect anomalous patterns in time-series sensor data, achieving 95.7% F1-score with sub-100ms latency.

Publications

Enhancement of various images using coefficients of Analytic Functions with Python™ 

International Conference on Intelligent Computing and Sustainable Innovation in Technology (IC-SIT) 2024

Courses and Certificates

Google Advanced Data Analytics 

Completed the Google Advanced Data Analytics program, gaining hands-on experience in statistics, exploratory data analysis (EDA), and machine learning. Built skills in Python-based data analysis and developed predictive modeling workflows.

Machine Learning and Data Science A-Z: Hands-on Python 2024 (Udemy) 

Gained practical experience with NumPy, Pandas, and Matplotlib for data analysis and visualization. Built various machine learning models using supervised and unsupervised learning techniques.

Database Management Systems (RDBMS) & Microsoft Fabric SQL 

Completed RDBMS & Microsoft Fabric SQL certification (Infosys Springboard) with practical exposure to relational schema design, normalization, and advanced SQL query optimization